MERIDIAN JUNIOR COLLEGE
JC2 Preliminary Examination 2018
Higher 1

CANDIDATE NAME

CIVICS GROUP | 1 | 7 | | | | |

INDEX NUMBER | | |

H1 BIOLOGY

Paper 2

ANSWER SCHEME

8876

12 September 2018

2 hours

READ THESE INSTRUCTIONS FIRST

Do not open this booklet until you are told to do so.
Write your name, civics group and index number on all the work you hand in.
Write in dark blue or black pen.
You may use a soft pencil for any diagrams or graphs.
Do not use staples, paper clips, glue or correction fluid.

Section A
Answer **all** questions in the spaces provided on the Question Paper.

Section B
Answer any **ONE** question in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.
You may lose marks if you do not show your working or if you do not use appropriate units.

At the end of the examination, fasten all your work securely together.
The number of marks is given in brackets [ ] at the end of each question or part question.

For examiner's Use	
Section A	
1	/ 9
2	/ 10
3	/ 11
4	/ 15
Section B	
5 or 6	/ 15
Total	/ 60

This document consists of **14** printed pages

[Turn over]

Section A

Answer **all** questions in this section

QUESTION 1

(a) Distinguish the processes of facilitated diffusion and active transport. [3]

- Facilitated diffusion involves transport of substances **down a concentration gradient** whereas active transport occurs **against a concentration gradient**.
- Facilitated diffusion **does not require ATP** whereas active transport **requires ATP**.
- Facilitated diffusion involves **channel or carrier proteins** whereas active transport only involves **protein pump**.
- .

(b) A group of students investigated the uptake of chloride ions in barley plants. They divided the plants into two groups and placed their roots in solutions containing radioactive chloride ions.

- Group **A** plants had a substance that inhibited respiration added to the solution.
- Group **B** did not have the substance added to the solution.

The students calculated the total amount of chloride ions absorbed by the plants every 15 minutes. Their results are shown in Fig. 1.1

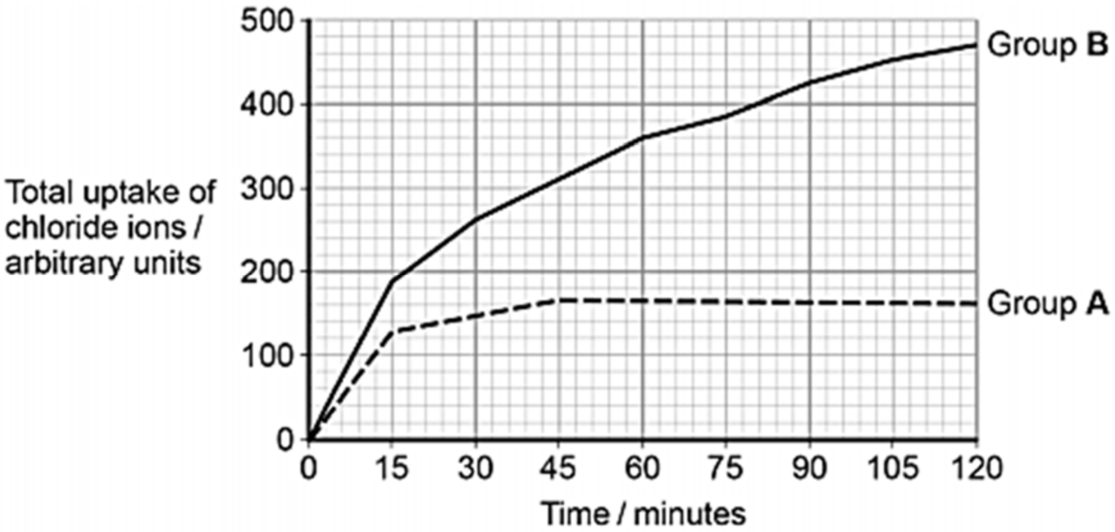

Fig. 1.1

(i) Calculate the ratio of the **rate** of uptake of chloride ions in the first hour to the **rate** of uptake of chloride ions in the second hour for group **B** plants. [2]

rate of uptake in the first hour = $\frac{360 - 0}{60}$

= 6 au min^{-1}

rate of uptake in the second hour = $\frac{470 - 360}{60}$

= 1.83 au min^{-1}

Ratio = 6/ 1.83 = **3.3 : 1**

(ii) Explain the results shown in Fig 1.1 [4]

- In Group A, **from 0-15 mins**, the initial rate of uptake of chloride ions is slower as only **facilitated diffusion** occurred.
- In Group A, **from 45-120 mins**, the total uptake of chloride ions **levels off/plateaus** because the concentrations of chloride ions inside cells and outside cells is the **same**/**reached equilibrium**.
- In Group B, **from 0-15 mins**, the initial rate of uptake of chloride ions is faster as both **facilitated diffusion** and **active transport** occurred.
- In Group B, **from 15-120 mins,** the total uptake of chloride ions continued to increase because the uptake of chloride ions is **against concentration/did not reach an equilibrium**.
- In Group B, **from 15-120 mins**, the rate of uptake **slows down** as **fewer chloride ions in external solution/ respiratory substrate is used up**.

[Total: 9]

QUESTION 2

Bone marrow contains many stem cells. Some of these stem cells are responsible for the replacement of white blood cells.

Fig. 2.1 shows the production of a white blood cell from one of these stem cells.

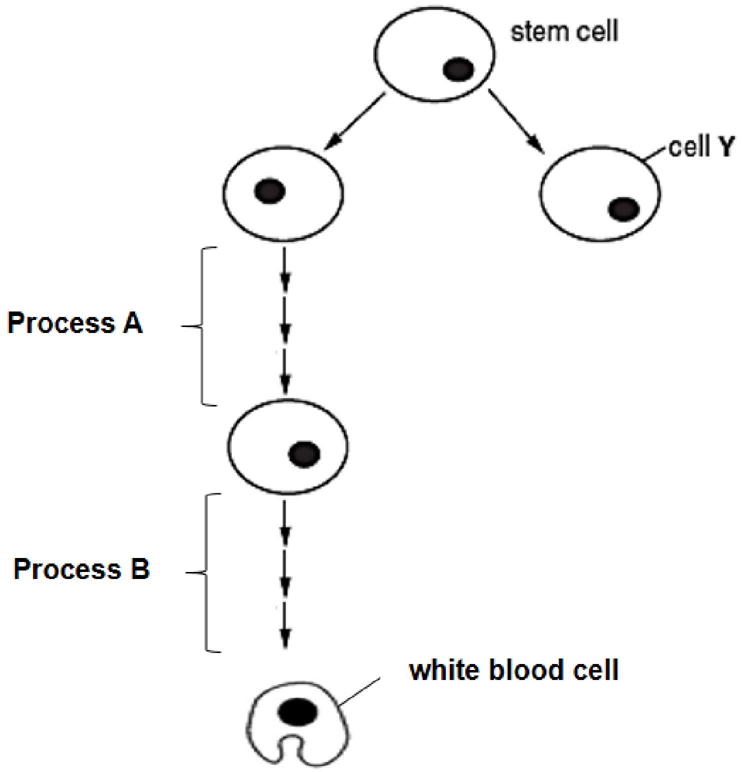

Fig. 2.1

(a) (i) State the potency of the stem cells found in the bone marrow. [1]

- Multipotent

(ii) Name the processes **A** and **B**. [2]

- **Process A – Mitosis**
- **Process B – Differentiation**

(iii) Suggest what may happen to cell **Y**. [1]

- Remains as a stem cell
- Undergoes cell division/mitosis
- Undergoes differentiation to become specialised cells

The human induced pluripotent stem cells (HiPS cells) can be derived from fully differentiated adult somatic cells such as white blood cells. This has been hailed as an effective replacement for human embryonic stem cells for its usefulness in regenerative medicine.

Recent research shows that abnormal reprogramming can occur during the induction of HiPS cells and tumours could be generated.

(b) (i) Compare the embryonic stem cell and HiPS cell. [3]

Difference

- Embryonic stem cell is obtained from **Inner cell mass of blastocyst** / **embryos** whereas HiPS cells are obtained from **differentiated somatic cell**

Similarities [any 2]

- Both are pluripotent /have the potential to become any cell type in the adult body but not those of the extra-embryonic membranes
- Both have **self-renewing capabilities** / can **divide continually**
- Both are **unspecialised/undifferentiated**

(ii) Comment on the ethical issues involved in the use of HiPS cells. [3]

- Overcome ethical issue of using embryonic stem cells for treatment as it does not involve killing of embryo.
- No tissue rejection involved as the HiPS cells are derived from the patient
- [idea of] Treatment to benefit patients/ alleviate the suffering of many people
- [idea of] Safety concerns on the use of HiPS cells.
- Potential to develop into a human embryo/ to clone human being/produce germ cells

[Total: 10]

QUESTION 3
In mice, fur colour is controlled by a gene with multiple alleles. These alleles are listed below in no particular order.

black and tan = C^{bt}	yellow = C^{y}
agouti = C^{a}	black = C^{b}

(a) Explain the term *multiple alleles*. [2]

- Gene that exists in **more than two allelic forms** in a given population
- **only two** of which can be present in a **diploid** organism

(b) Crosses between heterozygous parents with the genotype $C^{y}C^{b}$ always produce a ratio of two yellow mice to one black mouse. Explain the observation using a genetic diagram. [3]

Parental phenotype:	Yellow mouse	x	Yellow mouse		**[1]**
Parental genotype (2n):	C^{y} C^{b}	x	C^{y} C^{b}		
Gametes (n):	C^{y} C^{b}	x	C^{y} C^{b}		**[1]**

F_1 genotype (2n):	C^{y} C^{y}	C^{y} C^{b}	C^{y} C^{b}	C^{b} C^{b}	
F_1 phenotype:	mouse dies	yellow fur	yellow fur	black fur	**[1]**

F_1 phenotypic ratio: yellow fur : black fur
2 : 1

(c) Suggest explanations for the results of the following crosses between mice.

(i) Mice with agouti fur crossed with mice with black fur may produce all agouti offspring **or** some agouti and some black offspring. [2]

- Agouti allele (C^{a}) is **dominant** to black allele (C^{b})
- Parents with agouti fur has genotype of **C^{a} C^{a}** or **C^{a} C^{b}**
- Parents with black fur has genotype of **C^{b} C^{b}**

(ii) Mice with yellow fur crossed with mice with black fur will produce one of the following outcomes:

- some yellow offspring and some agouti offspring
- some yellow offspring and some black and tan offspring
- some yellow offspring and some black offspring. [2]

- yellow allele (C^{y}) is dominant to all alleles (C^{a}, C^{bt}, C^{b})
- Agouti allele (C^{a}) and black and tan allele are dominant to black allele (C^{b}) OR black allele is recessive to all other alleles
- Yellow mice is heterozygous at the gene locus

(d) A test cross is used to determine the genotype of an organism.

Describe how you would carry out a test cross to determine the genotype of a black and tan mouse. [2]

- Cross black and tan mouse with black mouse (C^b C^b)
- If **all** offspring is black and tan, then parent is homozygous (C^{bt} C^{bt})
- If some offspring are black and some are black and tan, then parent is heterozygous (C^{bt} _)

[Total: 11]

QUESTION 4

Malaria is a mosquito-borne infectious disease caused by the malarial parasites, *Plasmodium falciparum*, which is transmitted among humans by the female mosquito vectors *Anopheles*.

(a) Outline the general life cycle of a mosquito. [3]

- The female mosquitoes lay **eggs** in stagnant **water**
- develop as **larvae and pupae in water** [sequence must be correct]
- mature/develop into an **adult mosquito**.

Malaria was common in Italy, a European country situated in the Northern Hemisphere. Widespread land drainage together with the use of the insecticide DDT and the drug chloroquine eradicated both the mosquito vectors and the malarial parasites in the 1950s. Due to the success of these measures, they were later discontinued as they were no longer necessary.

Articles in the scientific literature more recently show that malarial mosquitoes are returning to Italy and increasing their numbers and their northerly range, with some cases of malaria being reported. In general, winters are milder and summers are hotter in the south of the country, with temperatures decreasing to the north, especially in the winters.

(b) Discuss whether the return of malaria to Italy can be attributed to climate change. [4]

[Yes]

- Range extending further north may relate to **warmer temperatures**
- [idea of] higher temperature lead to increase in number of mosquitoes due to e.g. hasten the life cycle of mosquitoes due to increased metabolism
- Climate change leads to **increase in rainfall** may result in **more flooded areas for mosquitoes to breed**

[No]

- Return of malaria was due to the **discontinued use of DDT**
- [idea of] Mosquitos could be introduced into Italy from the surrounding countries
- [idea of] The parasite thrive and spend most of their life cycle in host body temperature (37°C) hence ambient temperature has hardly any effect on their survival

(c) Chloroquine-resistant (CQR) malarial parasites, *P. falciparum,* were first reported in 1950s and are now widespread. The resistance is caused by mutations of a gene known as *pfcrt*.

(i) Explain why CQR *P. falciparum* are now widespread. [3]

- **Variation** in the population of *P. falciparum* due to random **mutations**
- Use of chloroquine acts as **selection pressure** on the *P. falciparum* population
- Chloroquine-resistant *P. falciparum* are at **selective advantage** and are likely to survive to reproduce
- ..and pass their **CQR allele/ mutated *pfcrt* allele** to their **fertile and viable offspring**
- **Allele frequency** for **CQR allele/ mutated *pfcrt* increases** over **many generations** within the *P. falciparum* population.

Analysis of *pfcrt* in CQR *P. falciparum* from different parts of the world shows differences in one section of the gene only. The amino acid sequences coded for by this section are shown in Table 4.1, together with the amino acid sequence coded for by the allele found in parasites that are still susceptible to chloroquine (non-CQR). The amino acid sequence coded for by the rest of the gene is the same in all cases.

Table 4.1

allele of *pfcrt* gene	amino acid sequence coded for by allele of *pfcrt* gene
non-CQR	—Cys—Met—Asn—Lys—His—Ala—Glu—Asn—Ile —Met—
CQR from Africa	—Cys—Ile —Glu—Thr—His—Ser—Glu—Ser—Ile —Ile —
CQR from Asia	—Cys—Ile —Glu—Thr—His—Ser—Glu—Ser—Thr—Ile —
CQR from Peru and Brazil	—Ser—Met—Asn—Thr—His—Ser—Gln—Asp—Leu—Arg—
CQR from Colombia	—Cys—Met—Glu—Thr—Gln—Ser—Gln—Asn—Ile —Thr—

(ii) With reference to Table 4.1, calculate the mean number of different amino acids coded for by CQR alleles in comparison with the non-CQR allele. Show your working.

[2]

$\frac{6 + 7 + 7 + 6}{4}$ [1]

= **6.5** [1]

mean number of different amino acids …………………..

In the non-CQR *P. falciparum*, chloroquine accumulates in the digestive vacuole of *P. falciparum* and interferes with the detoxification of haem in the host red blood cell. This results in parasite death.

Research shows that there is a point mutation in the *pfcrt* gene of CQR *P. falciparum* which leads to the formation of a chloroquine-resistance transporter located in the digestive vacuole membrane of CQR *P. falciparum*.

This mutation changes the amino acid from lysine to threonine at the binding site of the chloroquine-resistance transporter.

(iii) Using the information provided, suggest how this mutation of *pfcrt* gene increases the chloroquine resistance *in P. falciparum*. [3]

- **Base-pair substitution of a nucleotide at the 1st or 2nd nucleotide of the triplet** in the *pfcrt* gene of CQR *P. falciparum*
- ..**change in mRNA codon** that **codes for threonine** instead of **lysine**
- ..alters the **shape of the binding site** of pfcrt protein to become **complementary to chloroquine /** change the **charge of the binding site** in pfcrt protein to allow **binding to chloroquine**
- ..chloroquine are **transported out** of the digestive vacuole.

[Total: 15]

Section B

Answer **ONE** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.
Your answers should be illustrated by large, clearly labelled diagrams, were appropriate.
Your answers must be in continuous prose, where appropriate.
You answers must be set out in parts **(a)**, **(b)**, etc., as indicated in the question.

QUESTION 5

(a) Following cytokinesis, one of the daughter cells may not have a nucleolus. This cell is able to divide once more and then the new daughter cells die.

Explain how the cell is able to survive for one more cell division and suggest why the new daughter cells then die. [7]

[Function of nucleolus]

1. The nucleolus is a site where **rRNA genes** are located and **transcribed** to produce **rRNA**.
2. The nucleolus is also the **site of assembly** of rRNA and ribosomal proteins to form the large and small **subunits of the ribosomes**.
3. Cells **without nucleolus** are **unable to synthesize ribosomes** that are required for translation to take place.

[How the cell without a nucleolus is able to survive for one more cell division]

4. Though this daughter cell has no nucleolus, it **contains sufficient ribosomes** derived **from the parental cell** during cell division, but is unable to produce new ribosomes.
5. Hence this cell is still able to use the ribosomes to **make proteins** to carry out **cellular activities**, allowing it to undergo cell division once more.

[Why the new daughter cells then die]

6. The new daughter cells produced have **no nucleoli**.
7. [idea of] The amount of ribosomes derived from the parental cell is insufficient to meet the demand of the cell (new daughter cells have half the original number of ribosomes present in the abnormal daughter cell).
8. As **insufficient ribosomes** cannot synthesize sufficient proteins to drive cellular processes/meet cellular demands (e.g. respiration), the new daughter cells die.

QWC [1]
Processes communicated accurately with no ambiguity.

(b) The same parents may produce offspring that are different in appearance. Discuss how this is brought about. [8]

1. [Definition of a gene] A **gene** is a **discrete unit of hereditary information** consisting of **specific nucleotide sequence in DNA**

2. Genetic information is encoded in gene which **determines the phenotype**

Different gametes produced by same parents due to:

Germline mutation

3. Mutations occur in **germ cells** of the parents which produce gametes that can be **inherited by the offspring** during fertilization to produce different phenotype.

Meiosis

4. During **prophase I**, **crossing-over between non-sister chromatids of homologous chromosomes** occur. This results in exchange of alleles between corresponding gene loci.

5. During **metaphase I** and **anaphase I**, there is **independent assortment and segregation of homologous chromosomes** respectively. The resulting daughter cells are different as each has only one of the two homologous chromosomes.

6. During **metaphase II** and **anaphase II,** there is **independent assortment and segregation of non-identical sister chromatids respectively**.

7. These processes give rise to a large number of **different chromosome combinations** in the **gametes.**

Random fertilisation of gametes

8. **The random fusion of gametes** from two different parents during **fertilization** adds variation to the offspring/ results in different appearance in the offspring

Phenotypic variation due to environment

9. **Genes can interact with the environment** to give rise to phenotypic variation /different appearance of the offspring

QUESTION 6

(a) All cells, from bacteria to humans, express their genetic information via the central dogma of molecular biology which depicts the flow of genetic information from DNA to protein.

Outline the processes involved in the central dogma of molecular biology, and explain, using an example, why the central dogma may not always hold true. [7]

[Transcription] max 3

1. **General transcription factors** and **RNA polymerase** bind to the **promoter**
2. **One** of the DNA strands is used as the **template** to synthesize **mRNA**
3. Template read from **3' to 5'**, mRNA synthesized from **5' to 3'**
4. RNA polymerase add ribonucleotides by **complementary base pairing** via **hydrogen bonds**
5. Catalyzes formation of **phosphodiester bonds** between adjacent ribonucleotides.

[Translation] max 3

6. **Mature mRNA** used as a **template** to synthesize **polypeptide**
7. mRNA read from **5' to 3'**, three bases (**codon**) at a time
8. Ribosome peptidyl transferase catalyses **peptide bond** between adjacent amino acids
9. **Each codon** codes for **one amino acid**
10. Punctuated: **AUG start**, **UAA/UGA/UAG stop**

[Why central dogma is not always true] – any 1

11. Flow may be reverse, from **RNA to DNA**
    Ref. to HIV: (+)ssRNA reverse transcribed to ssDNA

12. Flow may **end at RNA**
    Ref. to tRNA / rRNA

QWC [1]
Central Dogma processes communicated accurately and to include one specific example

(b) The expression of genes gives rise to proteins that affect the biochemical reactions and other processes in organisms.

Discuss, with examples, the importance of specific shapes of proteins in organisms.

[8]

1. Active site of enzyme has specific shape that substrate can fit into
2. Via lock and key mechanism
3. *[Importance]* To form enzyme-substrate complex/ products important for metabolic pathways / increase the rate of reaction
4. *[Example]* any *enzyme and substrate*

5. DNA to fit into binding site of proteins (point 6-9, 16-19)
6. *[Importance]* Ref. to **DNA replication**
7. *[Example]* DNA polymerase binds to 3'OH group of existing nucleotide/ helicase binds to origin of replication/ topoisomerase binds to DNA to unzip the double-stranded
8. *[Importance]* Ref. to **transcription**
9. *[Example]* **transcription factor** binding to DNA / **RNA polymerase** binds to promoter

10. Binding of substances to transport proteins
11. *[Importance]* Allows for movement of substances across cell membrane
12. *[Example]* transmembrane protein e.g. Na+ channel, Na+K+ pump, glucose transporter etc

13. *[Example]* Haemoglobin is made up of **4 polypeptides** and their **haem groups** to form a specific conformation
14. *[Importance]* allows it to **bind to oxygen molecules** to form oxyhaemoglobin/ transport oxygen to all parts of the body
15. Reference to **cooperative binding**

16. *[Importance]* Ref. to **separation of sister chromatids during anaphase**
17. E.g. kinetochore to bind to centromere via complementary shape

18. *[Importance]* Ref. to **amino acid activation**
19. E.g. tRNA anticodon bind to the anticodon attachment site of amino-acyl tRNA synthetase

QWC [1]
Importance of specific shapes communicated accurately and to include at least two examples

End of Paper